

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Optics

L-T-P: 3-1-0

Credits: 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Geometrical Optics: Thick Lenses and Lens systems, Ray Tracing, Fiber Optics, Optical Imaging and Aberrations, Microscopes and Telescopes. (8)

Diffraction: The Huygens-Fresnel Principle, Kirchhoff's diffraction theory, Fraunhofer diffraction, Single/ Double Slit, Multiple Slit, Rectangular aperture and Airy Disc, Rayleigh Criterion and Resolving Powers of Optical instruments. Fresnel Diffraction: Fresnel half period zones, Circular aperture and Vibration spiral, Straight slit, Fresnel's integrals and Cornu spiral. Fresnel diffraction at a straight edge. (14)

Polarisation; Polarisation by reflection and refraction, Polarisation of scattered light. Dichroic polarisers. Birefringence, Uniaxial and Biaxial crystals, O-ray and E-ray, Velocity ellipsoids. Nicol prism, Wollaston and Rochon prisms. Retarders, Compensators, Optical activity, Optical modulators (8)

EM waves in anisotropic media, dielectric interfaces, Goos-Hanschen shift, waves in periodic layers and absorbing media (4)

Elementary ideas of Lasers and Holography. (8)

Reference Books:

1. Fundamentals of Optics: F. A. Jenkins and H. E. White

2. Optics: E. Hecht

3. Principles of Optics: M. Born and E. Wolf

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Geometrical Optics: Thick Lenses and Lens systems, Ray Tracing, Fiber Optics, Optical Imaging and Aberrations, Microscopes and Telescopes.	8
2.	Diffraction: The Huygens-Fresnel Principle, Kirchhoff's diffraction theory, Fraunhofer diffraction, Single/ Double Slit, Multiple Slit, Rectangular aperture and Airy Disc, Rayleigh Criterion and Resolving Powers of Optical instruments. Fresnel Diffraction: Fresnel half period zones, Circular aperture and Vibration spiral, Straight slit, Fresnel's integrals and Cornu spiral. Fresnel diffraction at a straight edge.	14
3.	Polarisation; Polarisation by reflection and refraction, Polarisation of scattered light. Dichroic	8

	polarisers. Birefringence, Uniaxial and Biaxial crystals, O-ray and E-ray, Velocity ellipsoids. Nicol prism, Wollaston and Rochon prisms. Retarders, Compensators, Optical activity, Optical modulators	
4.	EM waves in anisotropic media, dielectric interfaces, Goos-Hanschen shift, waves in periodic layers and absorbing media	4
5.	Elementary ideas of Lasers and Holography.	8
Total number of hours		42